



Classifying Asteroids Below the Detection Limit of a Single Image

Renato Pino, Korinna Bayer, Steve Jurado

Advisor: Prof. Andrew Connolly

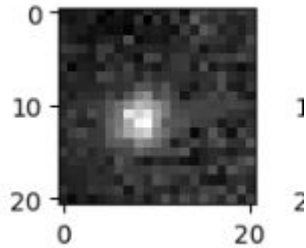
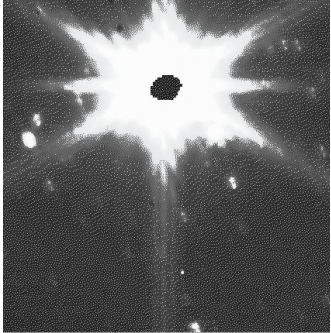
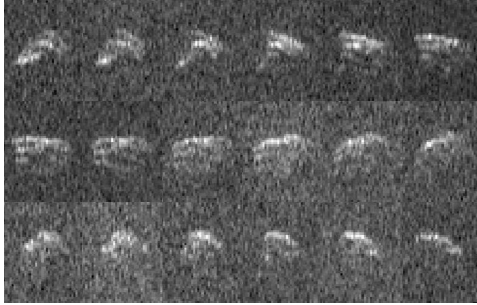
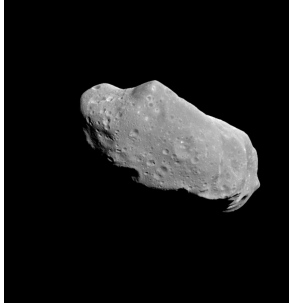


Astronomía

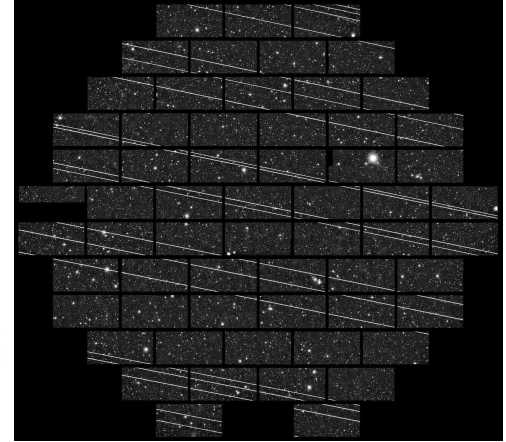
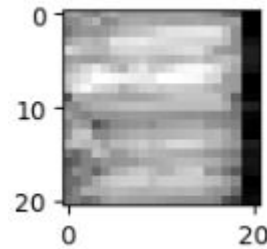
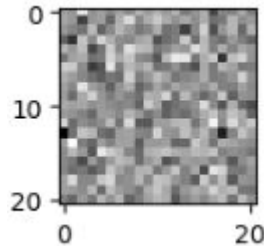
FACULTAD DE CIENCIAS
FÍSICAS Y MATEMÁTICAS
UNIVERSIDAD DE CHILE



What's That Object?

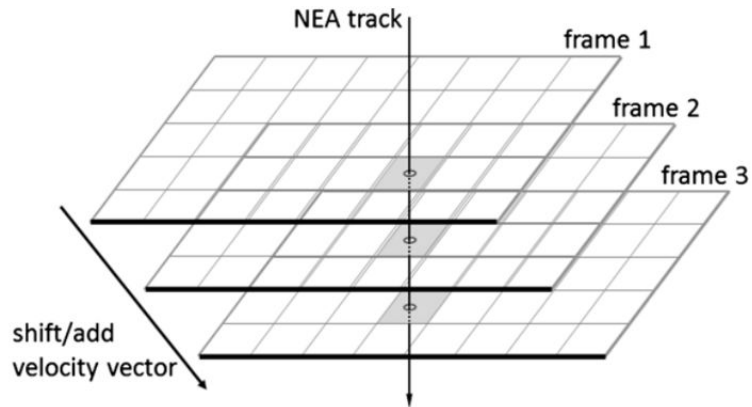
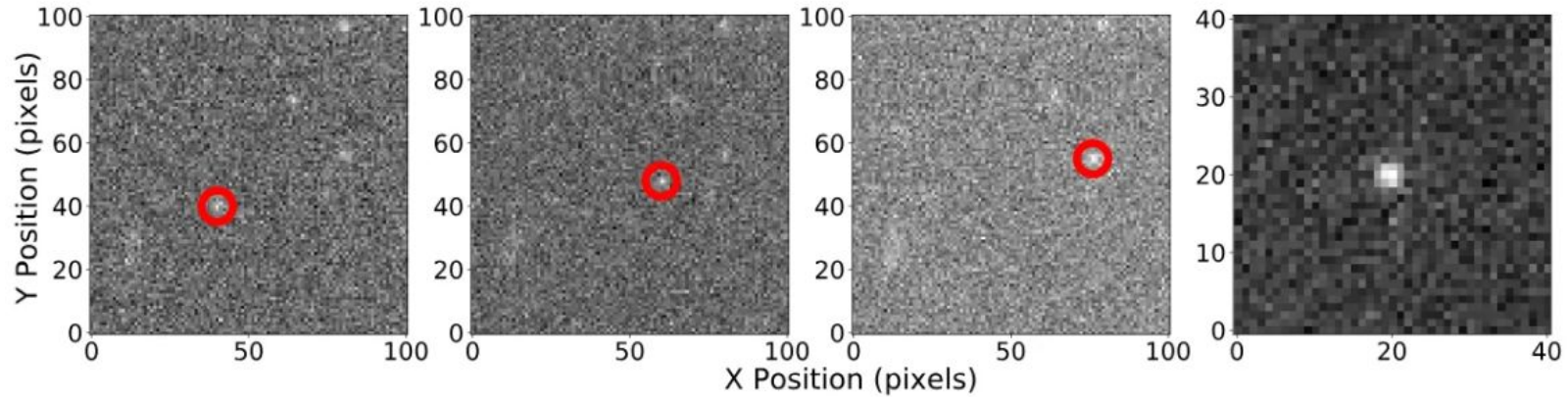


And this?



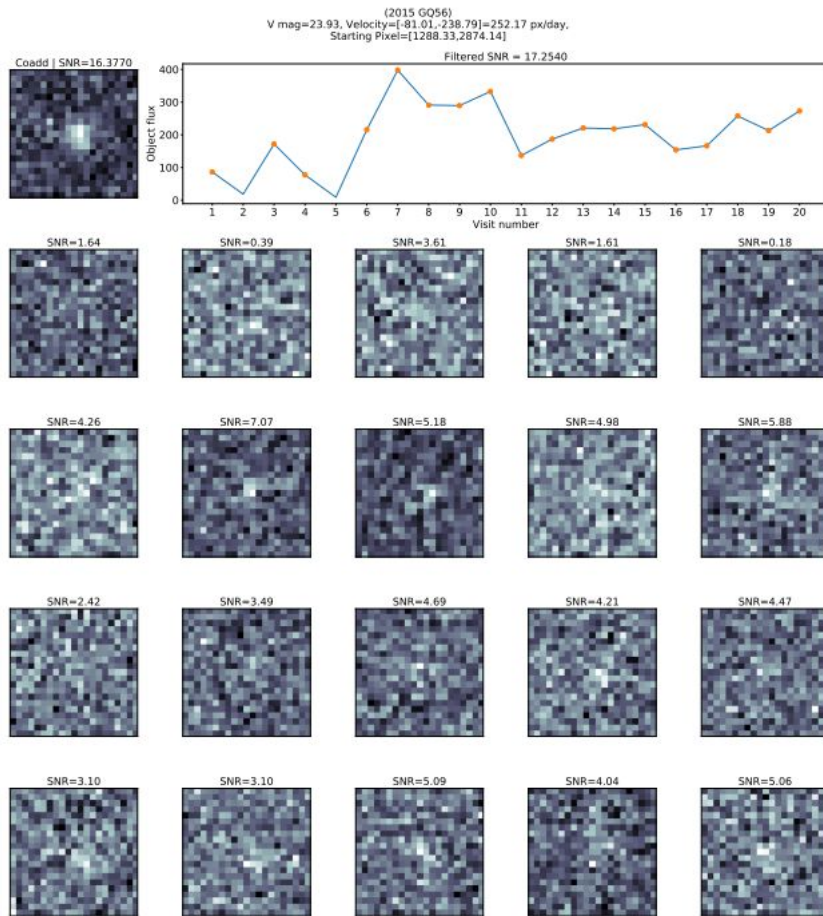
Asteroids

Others

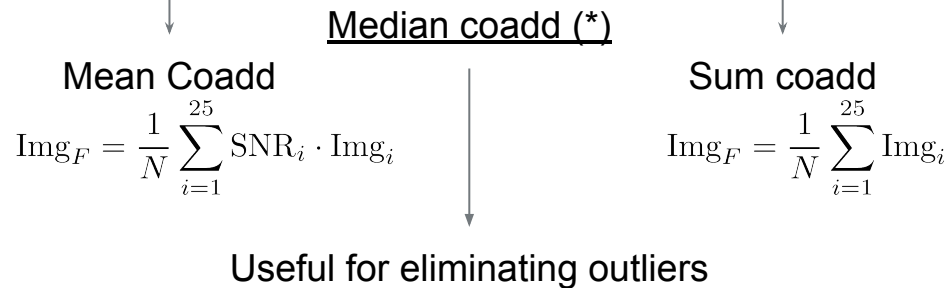


We can search for moving sources in stacks of images (e.g. planet 9 or populations of Trans-Neptunian Objects, TNOs)

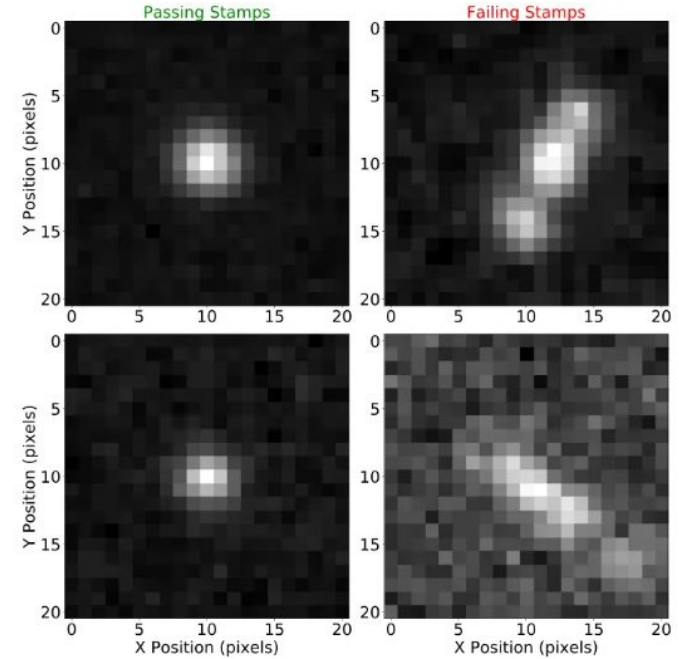
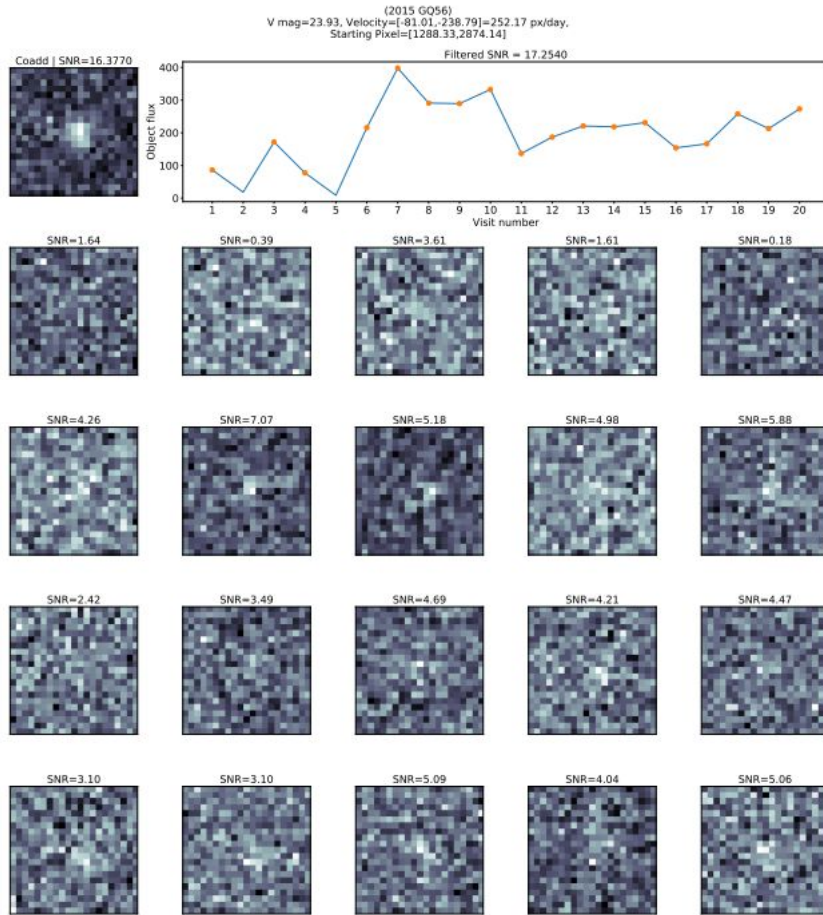
[Whidden et al. 2019](#)



Select a random subset of the total set to coadd, with a minimum number of observations set to 25

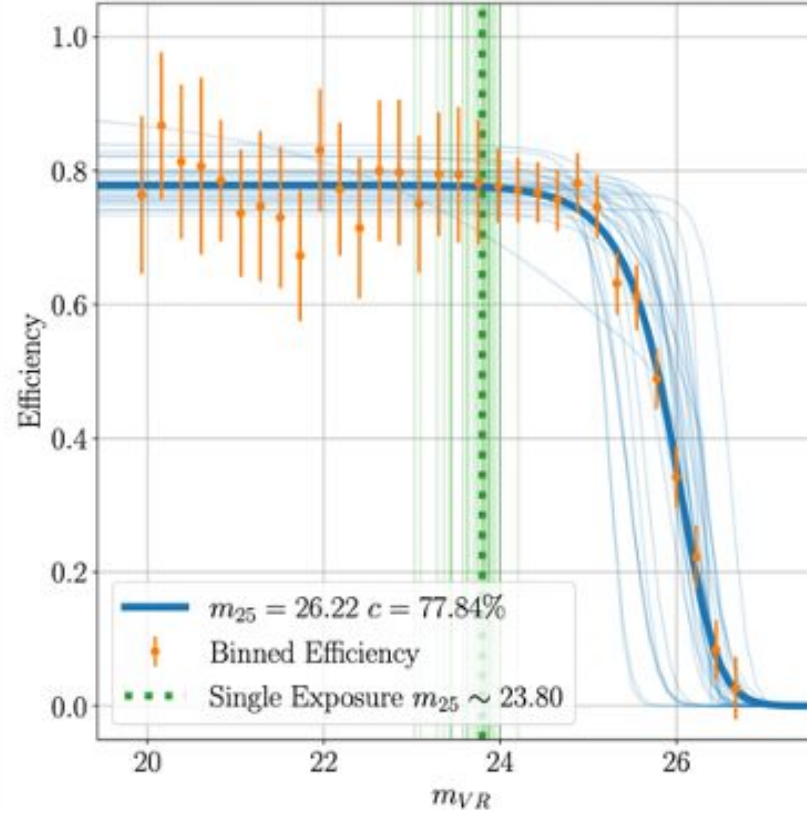


[Smotherman et al. 2023](#)



[Smotherman et al. 2023](#)

[Whidden et al. 2019](#)



[Smothernman et al. 2023](#)

Motivation

These are rare objects and searches are contaminated with many false positives (from ghosts and glints and artifacts in the data)

Objective

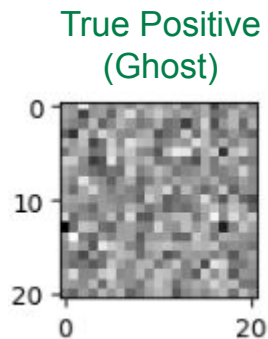
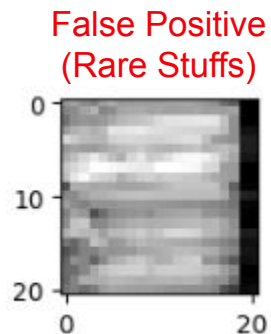
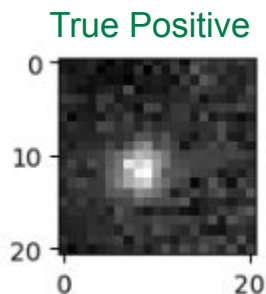
In this project we will try to identify the rare asteroids from the false positives using just a single image of faint asteroids.

To start by training a set of different CNNs to explore how well they classify the images.

Initially classify real asteroids from junk in the images.

As we progress we will move to using not just a single image but stacks of images where the asteroids are very faint.

In this project we will use images from the DEEP survey ([The DECam Ecliptic Exploration Project](#))



KBMOD uses GPU-accelerated computing to search over a wide range of trajectories for a stacks of CCDs in O(10) minutes, increasing the possible field of view of surveys to which digital tracking algorithms can be applied.

KBMOD is a set of python tools to search astronomical images for moving objects based upon method of maximum likelihood detection

$$n(\mathbf{x}, \mathbf{y}): \text{number of counts} \quad \Psi(\mathbf{y}) = \sum_i \frac{1}{\sigma_i^2} n(\mathbf{x}_i) T(\mathbf{x}_i - \mathbf{y}) \quad \Phi(\mathbf{y}) = \sum_i \frac{1}{\sigma_i^2} T(\mathbf{x}_i - \mathbf{y})^2 \quad T(\mathbf{x}, \mathbf{y}): \text{true sky intensity convolved with the PSF by the detector}$$

We make the assumption that the signal for the majority of the candidate detections will be dominated by the background noise and that the background noise is independent in each pixel. While there may be sources of correlation between pixels such as electronic detector effects (e.g., crosstalk or the dependence of the PSF on intensity), for this treatment we will assume that these effects are small enough to ignore and thus will continue as if we have a completely diagonal covariance.

where Ψ is the inverse-variance weighted cross-correlation of the PSF and the data, which is also the convolution of the PSF rotated by 180° or just the PSF if it is symmetric. Φ is the effective area of the PSF weighted by the inverse variance

$$\mathcal{L}_{\text{ML}}(\alpha_{\text{ML}}, \mathbf{y}) = \text{constant} - \frac{\Psi^2(\mathbf{y})}{2\Phi^2(\mathbf{y})} \quad \Phi_{\text{coadd}} = \sum_i \Phi_i(\mathbf{y}_i) \quad \Psi_{\text{coadd}} = \sum_i \Psi_i(\mathbf{y}_i)$$

The log form of a Gaussian— where by analogy Ψ ; $(\mathbf{x} - \mu)$ and Φ ; σ^2 —so we can approximate a χ^2 distribution. If we define $n = Y F$ and then make an image of v , we can call points above some threshold value m to be m -sigma detections

$$\nu_{\text{coadd}} = \frac{\Psi_{\text{coadd}}}{\sqrt{\Phi_{\text{coadd}}}} \quad \text{Signal-Noise}$$

Therefore, the final value for the likelihood of a point object is just the sum of the values at a given set of points in the Ψ image divided by the sum of the

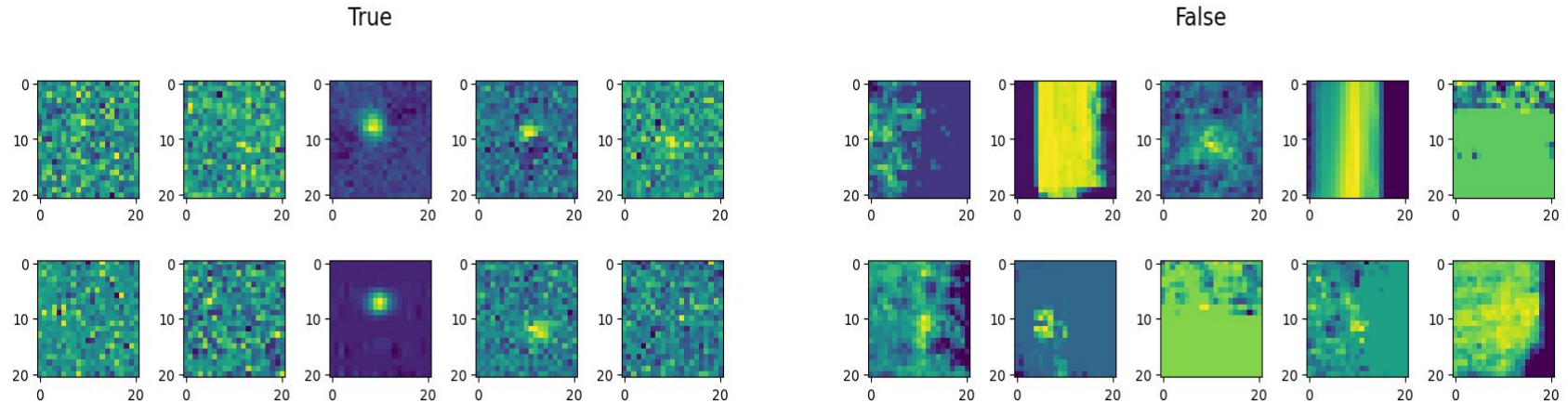
These are not FITS files for us

KBMOD is a set of python tools to search astronomical images for moving objects based upon method of maximum likelihood detection

DEEP survey (DECam Ecliptic Exploration Project) images.

Data set divided into true positives and false positives

- True positives: Synthetic objects inserted into DEEP data
- False positives: Running KBMOD on the DEEP data, but setting the set of the search angles to be roughly perpendicular to the ecliptic.



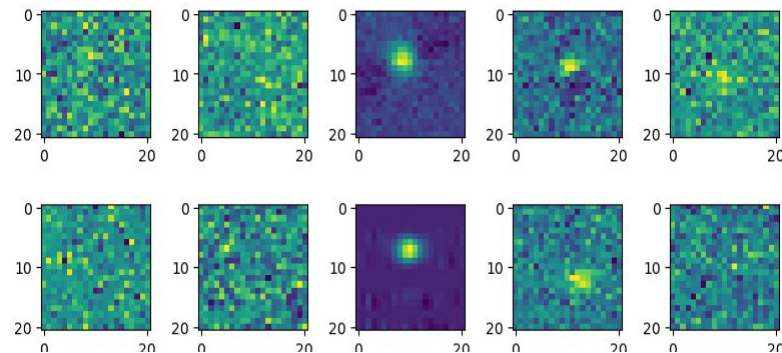
Approx 50:50 in each dataset

Type Dataset		# Coadd Images	
<u>TRAIN</u>	True Positives	129010	239540
	False Positives	110530	
<u>VALIDATION</u>	True Positives	36860	68440
	False Positives	31580	
<u>TEST</u>	True Positives	18430	34220
	False Positives	15790	

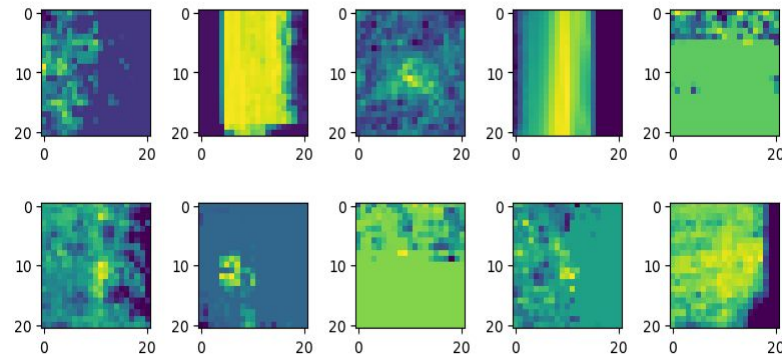
1 Channel, 21 x 21

342200

True



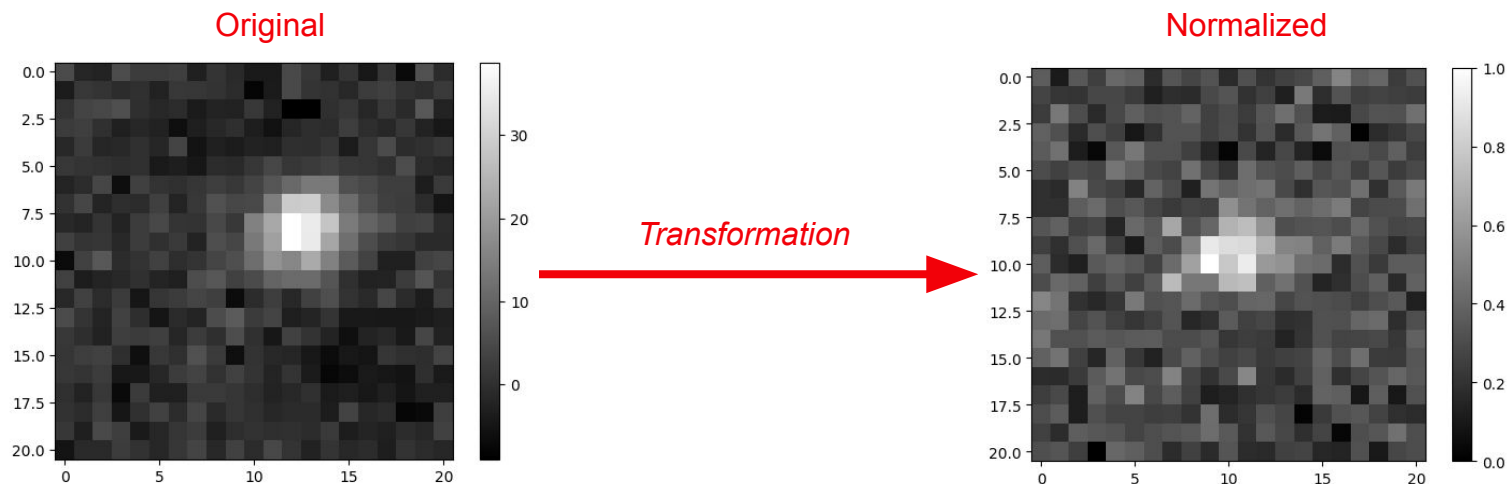
False



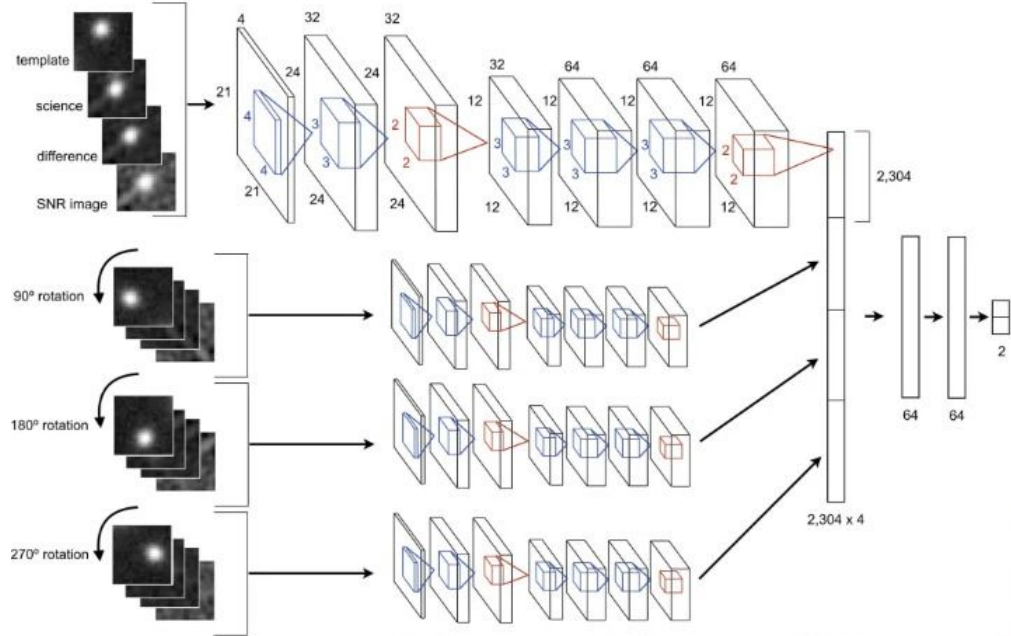
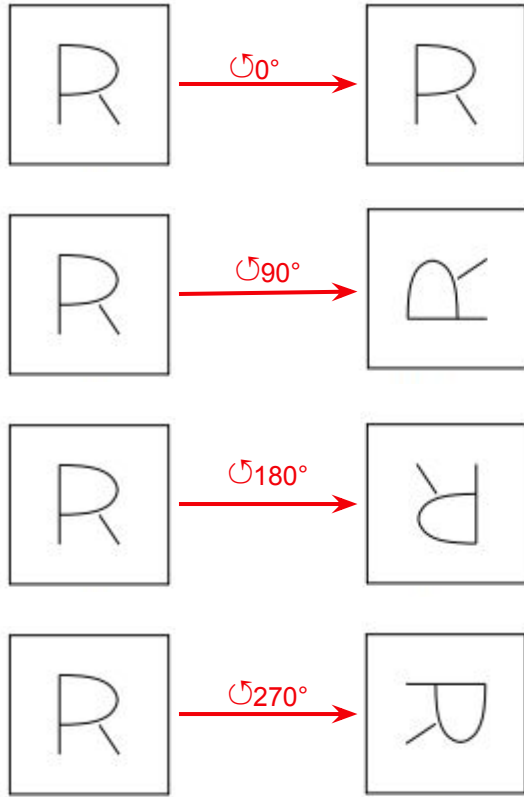
$$I_{\text{new}} = (I - I_{\text{min}}) \frac{I_{\text{new,max}} - I_{\text{new,min}}}{I_{\text{max}} - I_{\text{min}}} + I_{\text{new,min}}$$

Note: Red arrows in the original image point from $I_{\text{new,max}}$ to 1, from $I_{\text{new,min}}$ to 0, and from the final $I_{\text{new,min}}$ to 0.

$$I_{\text{new}} = \frac{I - I_{\text{min}}}{I_{\text{max}} - I_{\text{min}}} \quad I_{\text{new}} \in [0, 1]$$

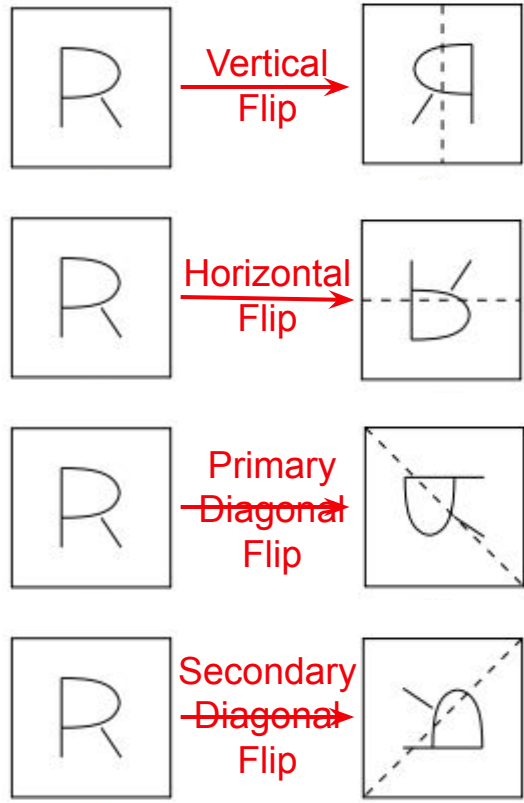
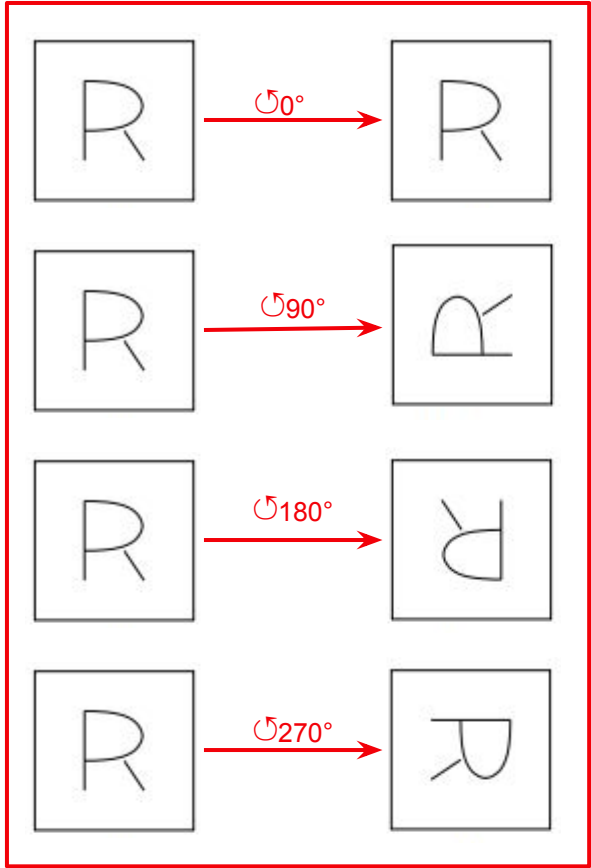


Data: Augmentation



[Cabrera-Vives et al. 2017](#)

Not complete!!!

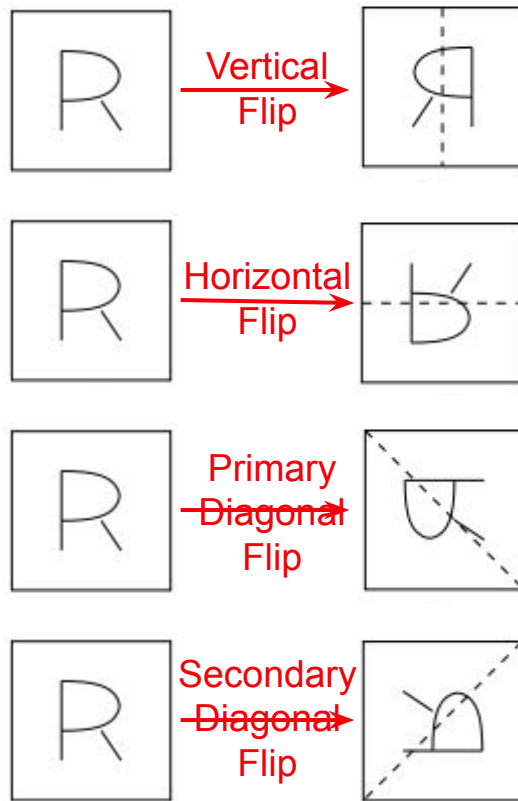
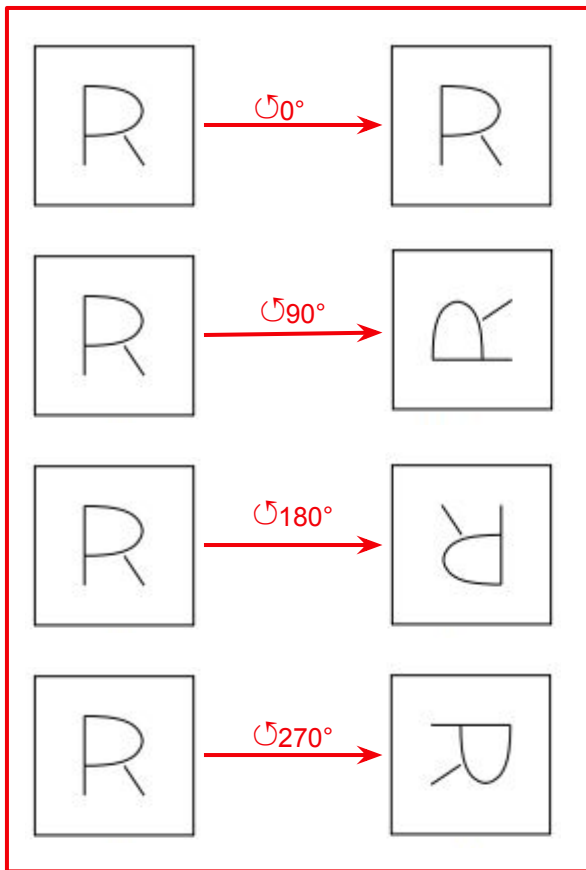


D4 Symmetric Group

	R_0	R_{90}	R_{180}	R_{270}
R_0	R_0	R_{90}	R_{180}	R_{270}
R_{90}	R_{90}	R_{180}	R_{270}	R_0
R_{180}	R_{180}	R_{270}	R_0	R_{90}
R_{270}	R_{270}	R_0	R_{90}	R_{180}
H	H	D	V	D'
V	V	D'	H	D
D	D	V	D'	H
D'	D'	H	D	V

Completed table

Data: Augmentation



D4
Symmetric Group

*Very heavy to process with
many images*

x8
Coadd Images

Training set preparation (same for Validation set and Test set):

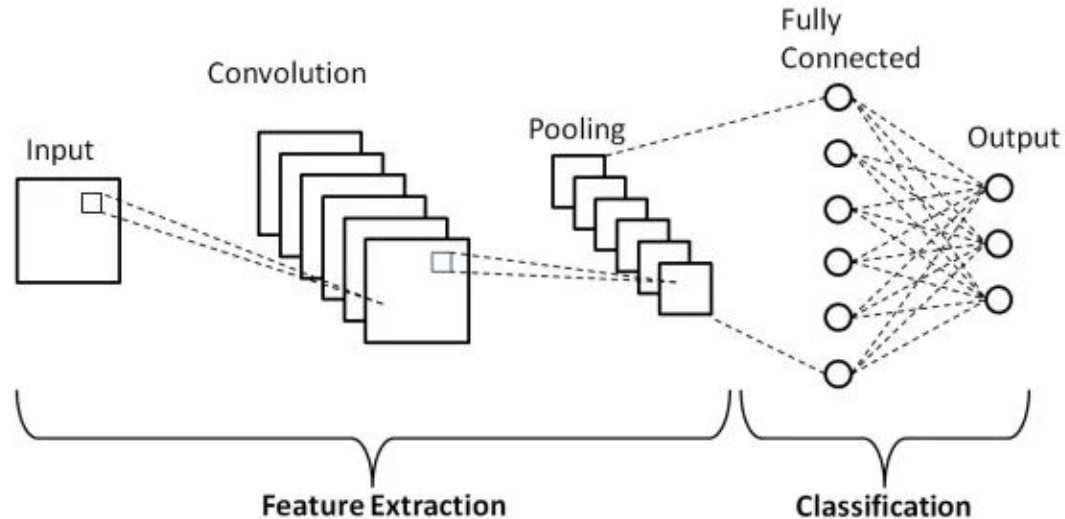
- Selection → Use of Median Coadd
- Concatenation → Vertically Stack: True Positives and False Positives into a single array.
- Label Assignment → Label Assignment: 0 for False Positives and 1 for True Positives.
- Shuffling
- One-hot encoding → [1, 0] for False Positives and [0, 1] for True Positives.
- Normalization.

Just for Training set → Augmentation.

Modeling

Since this is an image classification problem, we use convolutional neural networks.

We first test a simple CNN to get a baseline and then move forward using pre-trained models from ImageNet.



Loss Function	Optimizer	Batch Size	Learning Rate
Categorical CrossEntropy	Adam	512	0.0001

Simple or Basic CNN

```
nn.Conv2d(1, 8, kernel = (3,3)),  
nn.ReLU(),  
nn.BatchNorm2d(8),  
nn.MaxPool2d(kernel = (2,2),  
nn.Dropout(p=0.25),  
nn.Flatten(),  
nn.Linear(8*9*9, 64),
```

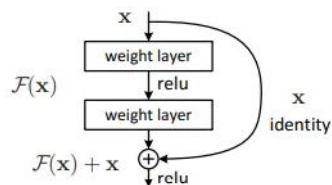
Complex CNN model

3 x Convolutional Layer, Batch
Normalization, Max Pooling
Layer, Connected Layer,
Dropout, Connected Layer,
Output Layer

Models: Features

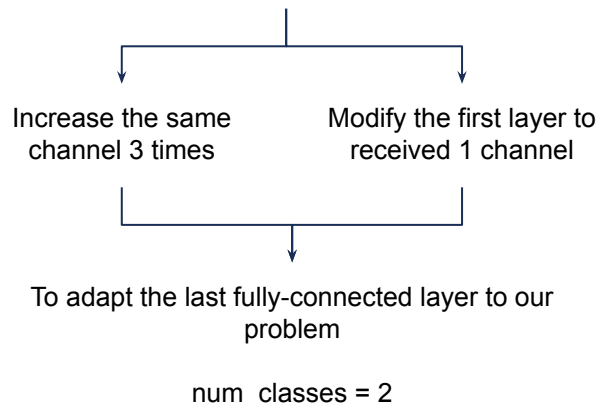
Loss Function	Optimizer	Batch Size	Learning Rate
Categorical CrossEntropy	Adam	512	0.0001

ResNet50 & ResNet56



layer name	output size	18-layer	34-layer	50-layer	101-layer	152-layer
conv1	112×112	7×7, 64, stride 2				
conv2_x	56×56	3×3 max pool, stride 2				
		$\begin{bmatrix} 3 \times 3, 64 \\ 3 \times 3, 64 \end{bmatrix} \times 2$	$\begin{bmatrix} 3 \times 3, 64 \\ 3 \times 3, 64 \end{bmatrix} \times 3$	$\begin{bmatrix} 1 \times 1, 64 \\ 3 \times 3, 64 \\ 1 \times 1, 256 \end{bmatrix} \times 3$	$\begin{bmatrix} 1 \times 1, 64 \\ 3 \times 3, 64 \\ 1 \times 1, 256 \end{bmatrix} \times 3$	$\begin{bmatrix} 1 \times 1, 64 \\ 3 \times 3, 64 \\ 1 \times 1, 256 \end{bmatrix} \times 3$
conv3_x	28×28	$\begin{bmatrix} 3 \times 3, 128 \\ 3 \times 3, 128 \end{bmatrix} \times 2$	$\begin{bmatrix} 3 \times 3, 128 \\ 3 \times 3, 128 \end{bmatrix} \times 4$	$\begin{bmatrix} 1 \times 1, 128 \\ 3 \times 3, 128 \\ 1 \times 1, 512 \end{bmatrix} \times 4$	$\begin{bmatrix} 1 \times 1, 128 \\ 3 \times 3, 128 \\ 1 \times 1, 512 \end{bmatrix} \times 4$	$\begin{bmatrix} 1 \times 1, 128 \\ 3 \times 3, 128 \\ 1 \times 1, 512 \end{bmatrix} \times 8$
conv4_x	14×14	$\begin{bmatrix} 3 \times 3, 256 \\ 3 \times 3, 256 \end{bmatrix} \times 2$	$\begin{bmatrix} 3 \times 3, 256 \\ 3 \times 3, 256 \end{bmatrix} \times 6$	$\begin{bmatrix} 1 \times 1, 256 \\ 3 \times 3, 256 \\ 1 \times 1, 1024 \end{bmatrix} \times 6$	$\begin{bmatrix} 1 \times 1, 256 \\ 3 \times 3, 256 \\ 1 \times 1, 1024 \end{bmatrix} \times 23$	$\begin{bmatrix} 1 \times 1, 256 \\ 3 \times 3, 256 \\ 1 \times 1, 1024 \end{bmatrix} \times 36$
conv5_x	7×7	$\begin{bmatrix} 3 \times 3, 512 \\ 3 \times 3, 512 \end{bmatrix} \times 2$	$\begin{bmatrix} 3 \times 3, 512 \\ 3 \times 3, 512 \end{bmatrix} \times 3$	$\begin{bmatrix} 1 \times 1, 512 \\ 3 \times 3, 512 \\ 1 \times 1, 2048 \end{bmatrix} \times 3$	$\begin{bmatrix} 1 \times 1, 512 \\ 3 \times 3, 512 \\ 1 \times 1, 2048 \end{bmatrix} \times 3$	$\begin{bmatrix} 1 \times 1, 512 \\ 3 \times 3, 512 \\ 1 \times 1, 2048 \end{bmatrix} \times 3$
	1×1	average pool, 1000-d fc, softmax				
FLOPs		1.8×10^9	3.6×10^9	3.8×10^9	7.6×10^9	11.3×10^9

Pretrained



[He et al. \(2015\)](#)

Modeling

Basic Model

```
=====
Total params: 41,762
Trainable params: 41,762
Non-trainable params: 0
-----
```

ResNet56

```
=====
Total params: 857,090
Trainable params: 857,090
Non-trainable params: 0
-----
```

ResNet50

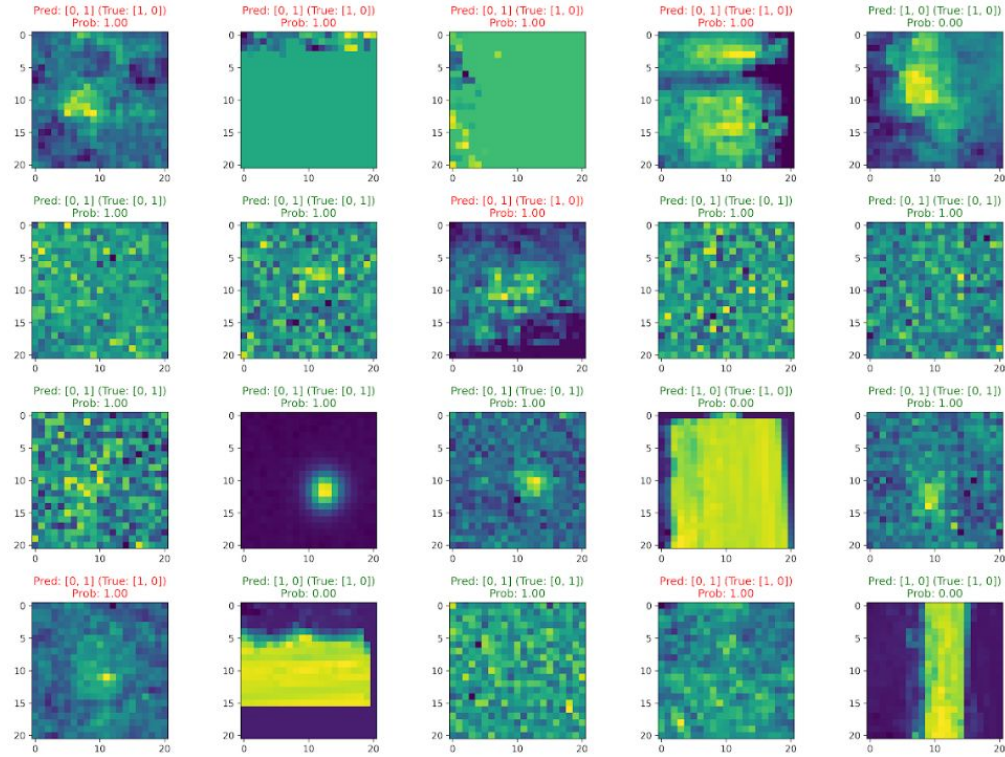
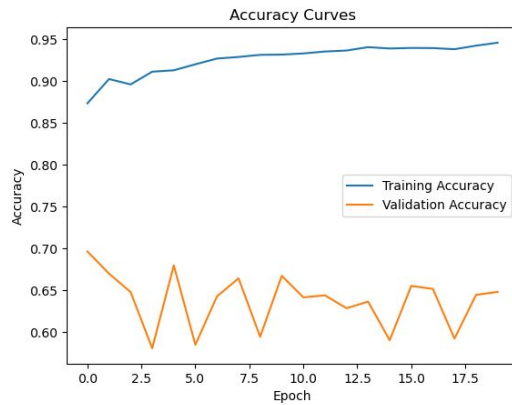
```
=====
Total params: 23,524,738
Trainable params: 23,524,738
Non-trainable params: 0
-----
```

Transfer Learning

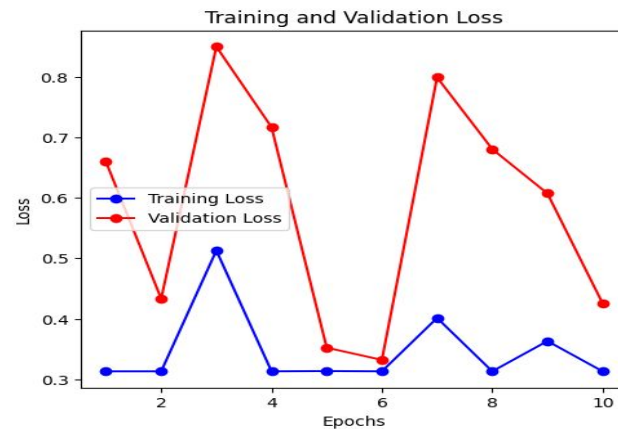
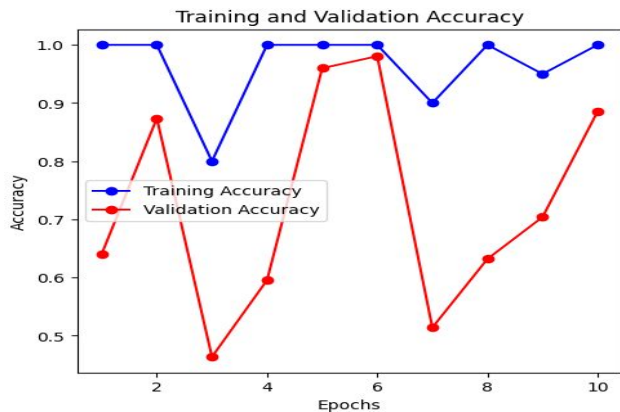
ResNet50

```
=====
Total params: 23,505,858
Trainable params: 7,234
Non-trainable params: 23,498,624
-----
```

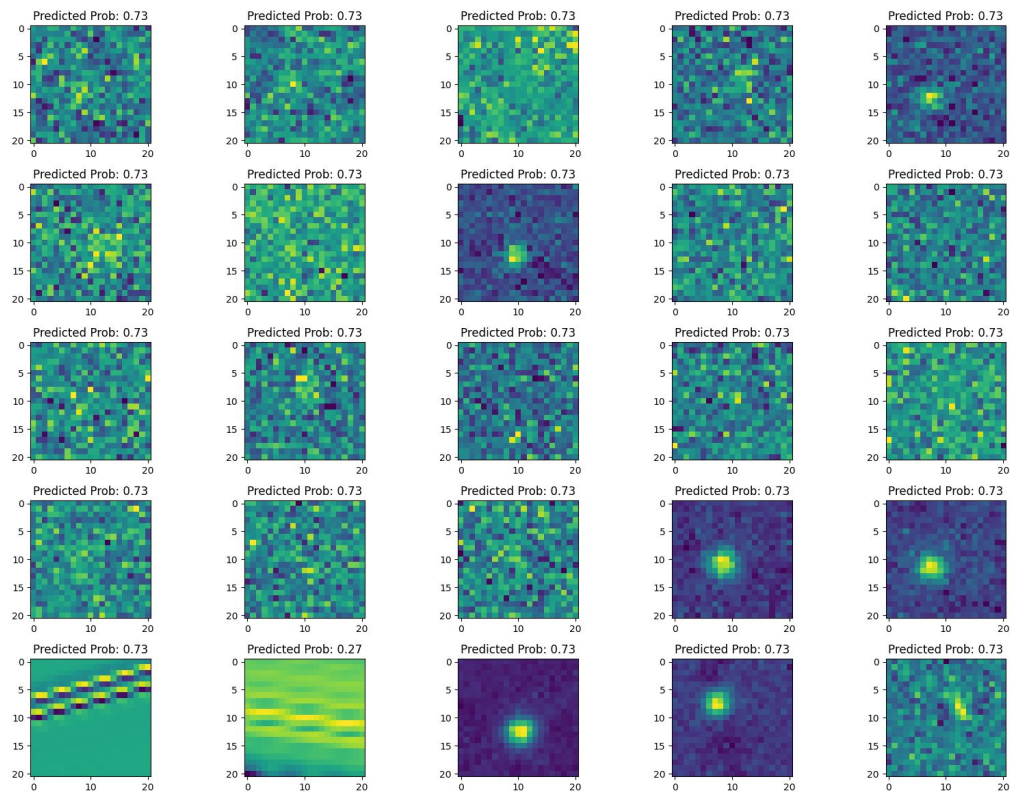

Cerro Calán
Observatorio Astronómico Nacional



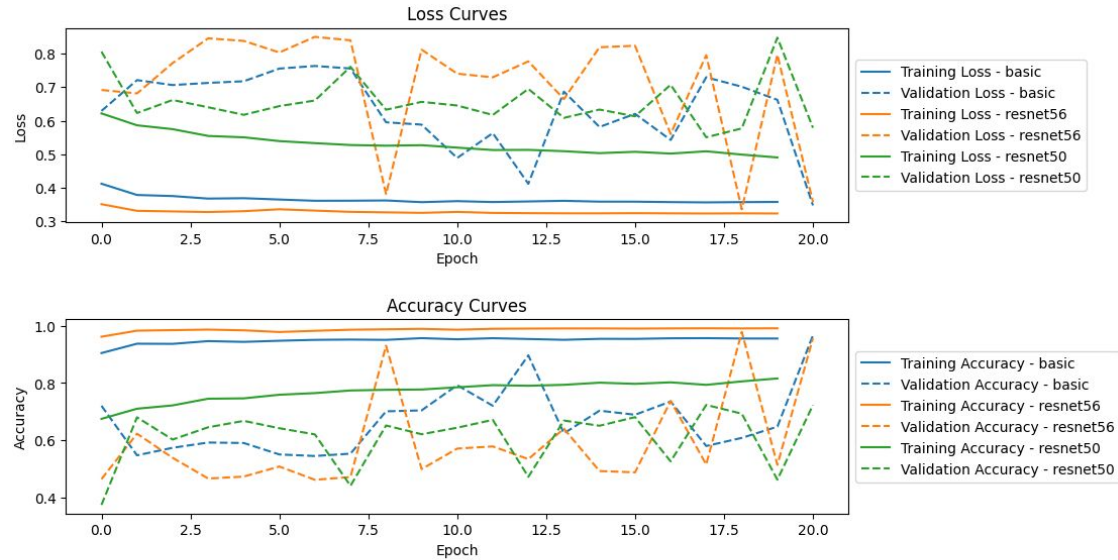
Complex CNN Results



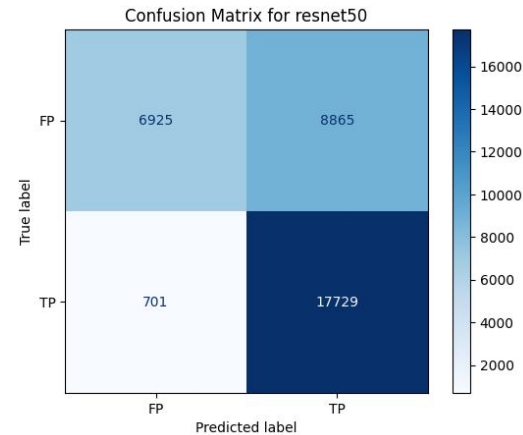
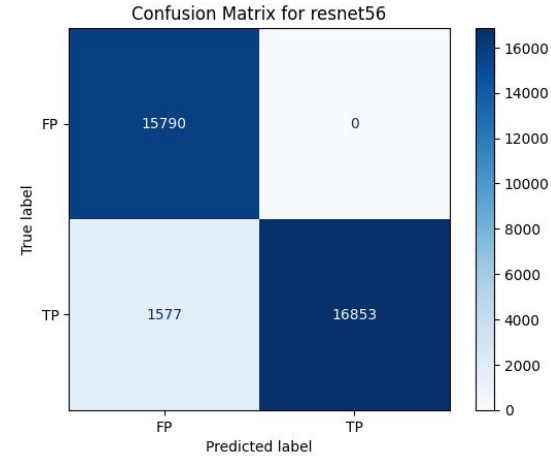
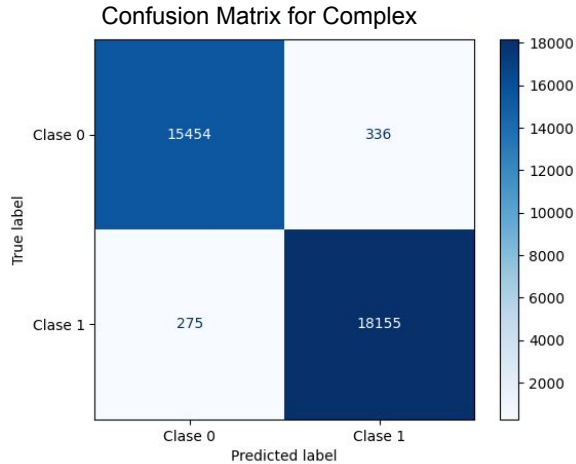
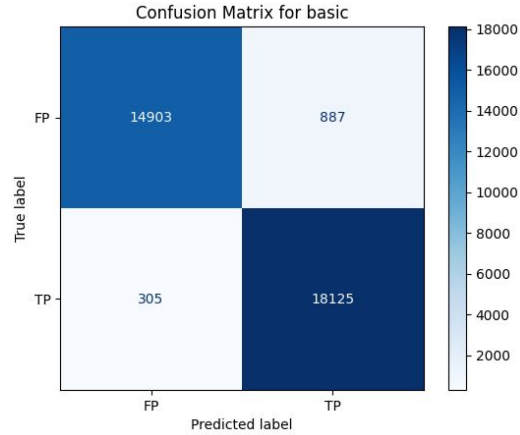
Real positives probabilities:



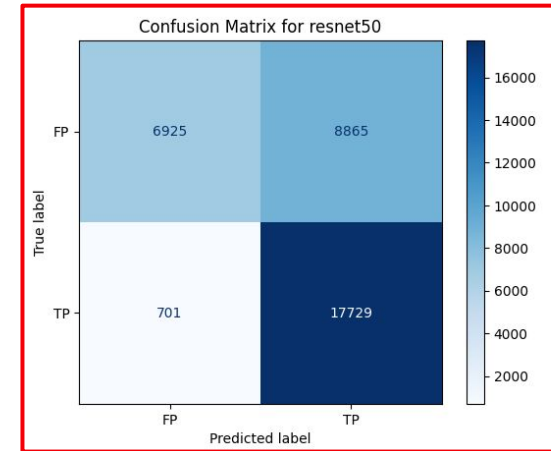
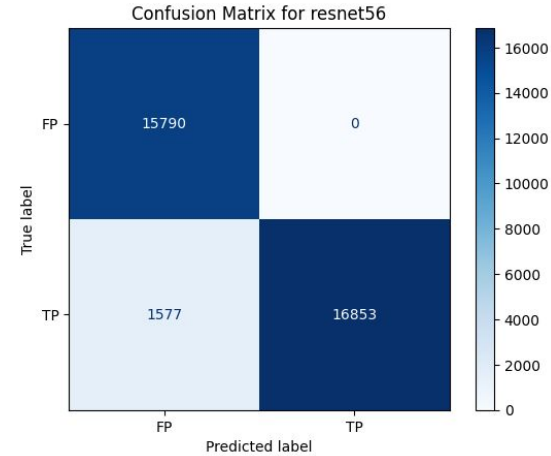
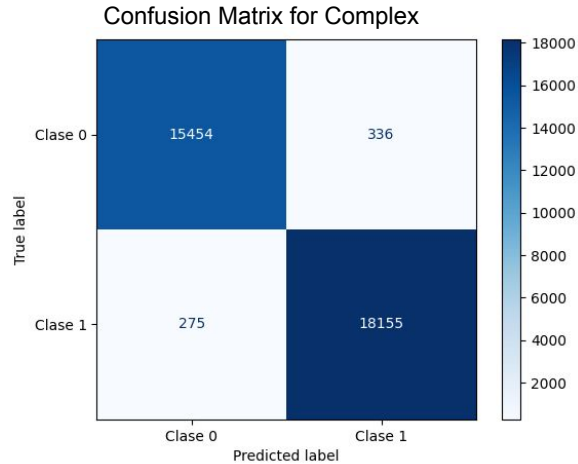
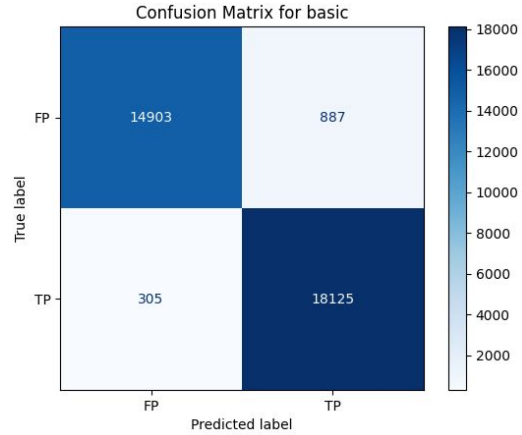
Results: Previous



Results



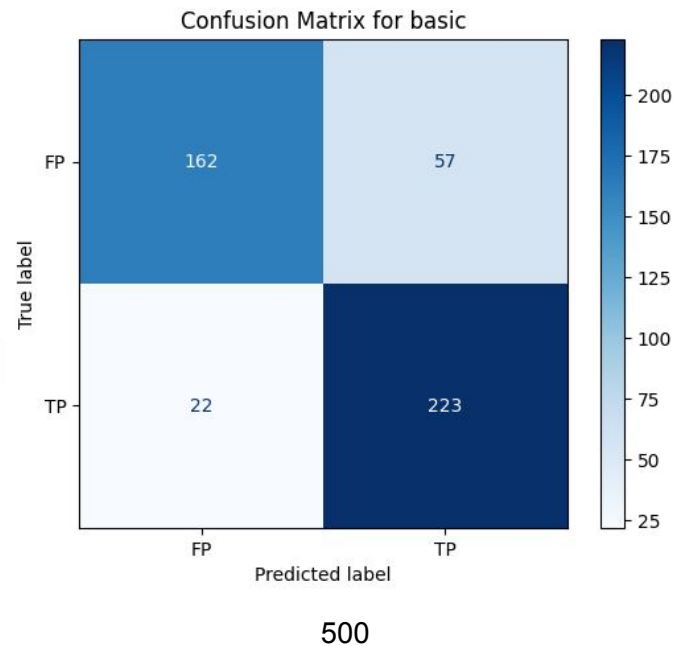
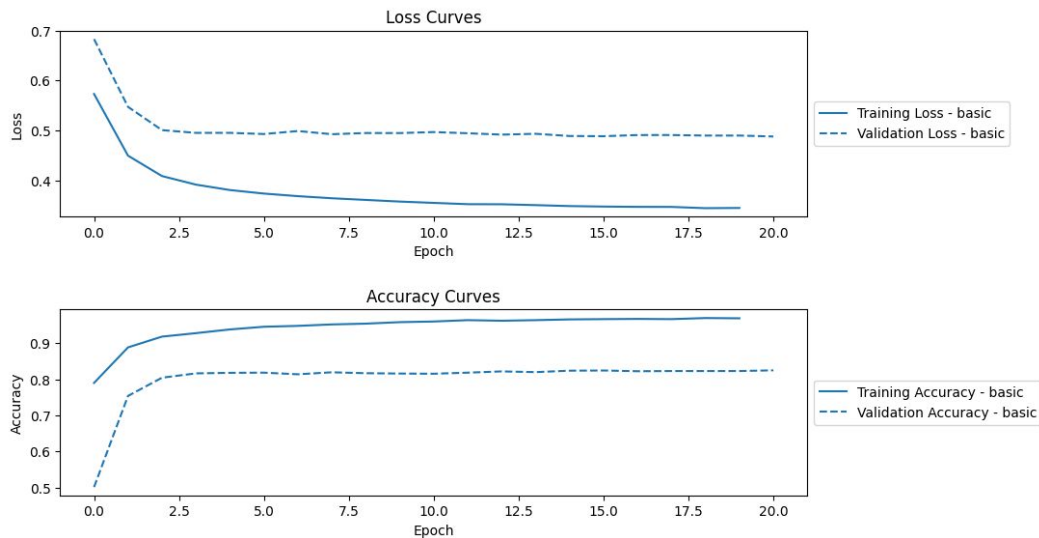
Results



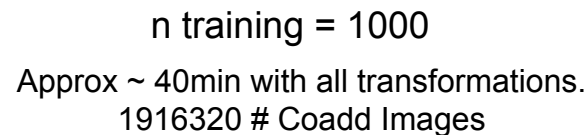
Results: Annex 8 transformations

n training = 1000

Approx ~ 40min with all transformations.
1916320 # Coadd Images



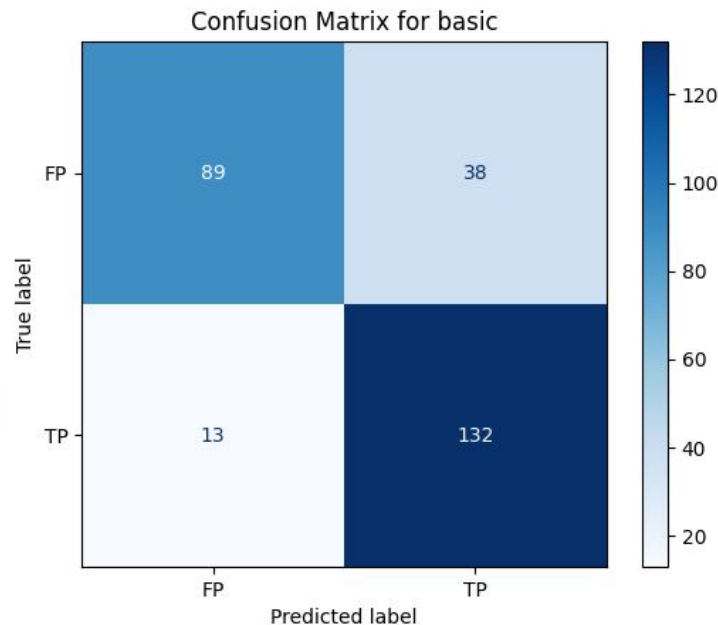
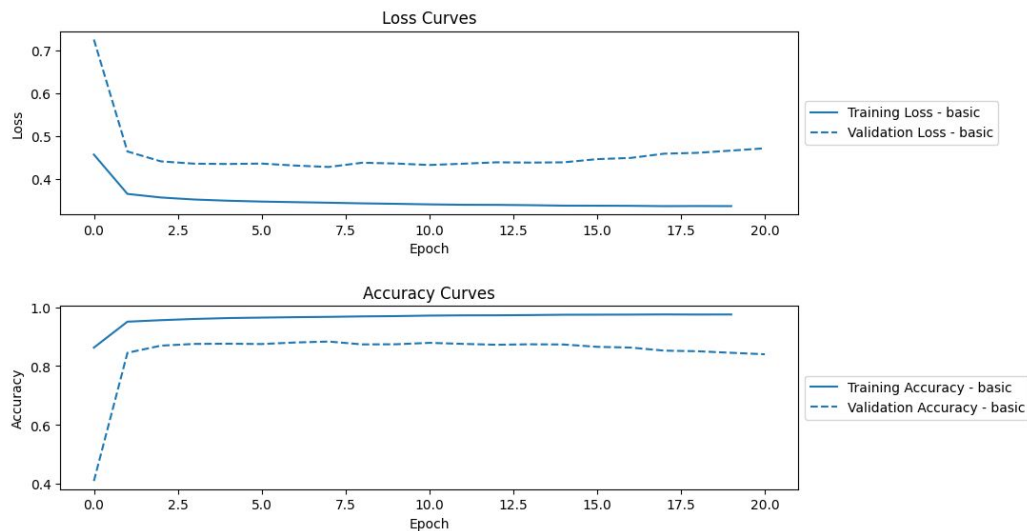
Cerro Calán
Observatorio Astronómico Nacional



Results: Annex 8 transformations

n training = 10000

Approx ~ 40min with all transformations.
1916320 # Coadd Images



Overfitting ?
Problem in the model ?

- Increasingly complicated models.
- To apply transformation as part of NN and not increase the input amount of images ([Förster et al. \(2022\)](#) [DELIGHT]).
- To apply the complex models over the increased input dataset
- Data transformations:
 - Filter out noisy images from true and false positives.
- To apply interpretability methods to images (e.g., SHAP explainer).
- To add astronomical information to more context.
- Explore other forms to represent images or even increases the channel dimensions.
- To use the mean cosas images and sum cold images

Cabrera-Vives, G. et al. (2017). Deep-HITS: Rotation Invariant Convolutional Neural Network for Transient Detection*. The Astrophysical Journal, 836(1), 97. <https://doi.org/10.3847/1538-4357/836/1/97>.

He, K., Zhang, X., Ren, S., & Sun, J. (2015, December 10). Deep Residual Learning for Image Recognition. ArXiv.org; arXiv. <https://arxiv.org/abs/1512.03385>.

Smotherman, H. et al. (2024). The DECAM Ecliptic Exploration Project (DEEP). VI. First multiyear observations of Trans-Neptunian objects. The Astronomical Journal, 167(3), 136. <https://doi.org/10.3847/1538-3881/ad1524>.

Whidden, P. J. et al. (2019). Fast Algorithms for Slow Moving Asteroids: Constraints on the Distribution of Kuiper Belt Objects. The Astronomical Journal, 157(3), 119. <https://doi.org/10.3847/1538-3881/aafd2d>.

Results

